Name: ____________________

Worksheet

Calculating theoretical probabilities from probability tree diagrams (one event)

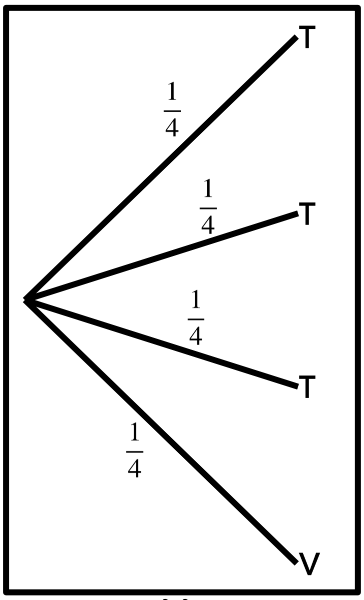
Task A: Probability trees to show outcomes

1) Both these **probability trees** show the outcomes and probabilities for the same trial (spinning a spinner).

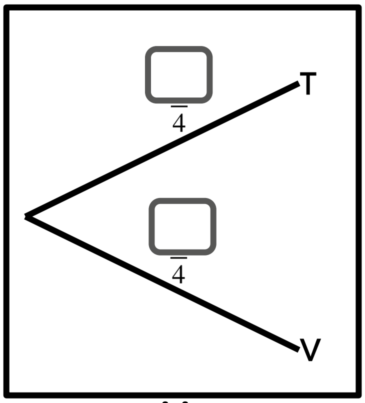
a) Use **probability tree** 1 to complete **probability tree** 2.

b) Use these two **probability trees** to complete the following sentence:

"The probability of the spinner landing on a T is ____________ ".



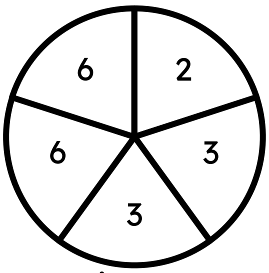
probability tree 1



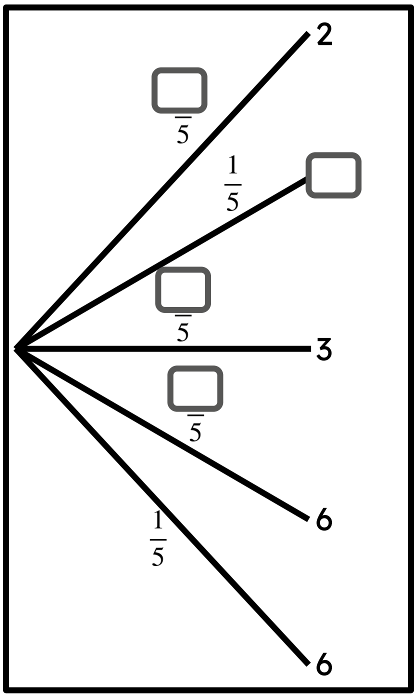
probability tree 2

2) Sofia will spin spinner A once.

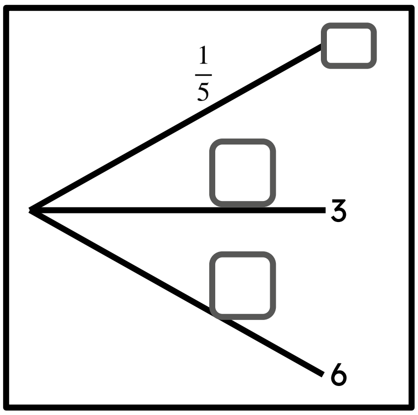
a) Complete the **probability tree** with five branches for this trial.



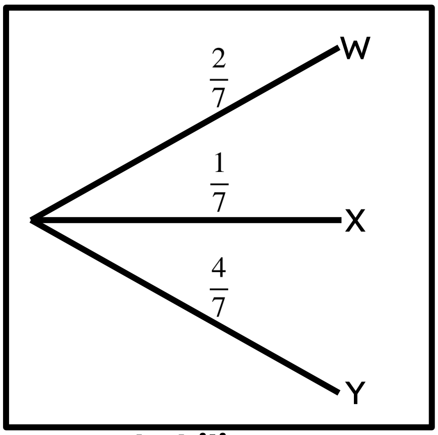
spinner A



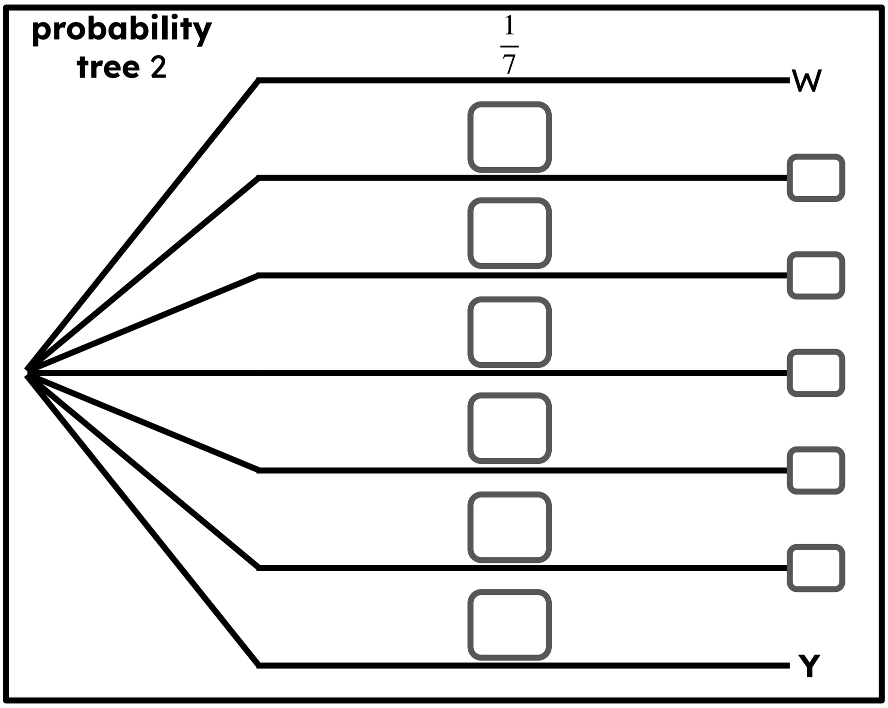
b) Hence, complete the **probability tree** which shows each of the unique outcomes for this trial.



3) Use **probability tree** 1 to complete **probability tree** 2.

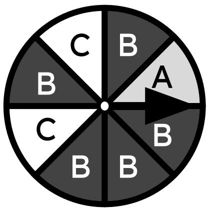


probability tree 1



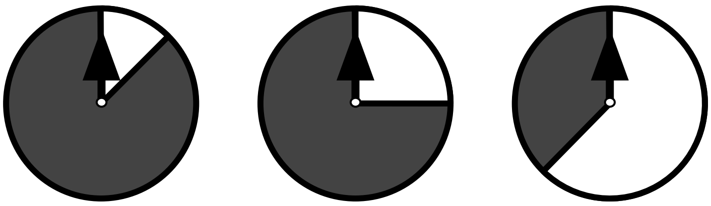
4) Jun will spin this spinner once.

a) By constructing an appropriate **probability tree**, list the probabilities of each unique outcome.



spinner

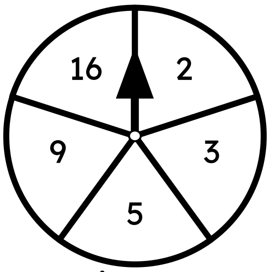
b) Mark the lighter segment on each of these outcome spinners with either A, B, or C to represent the probability that the spinner lands on that outcome.



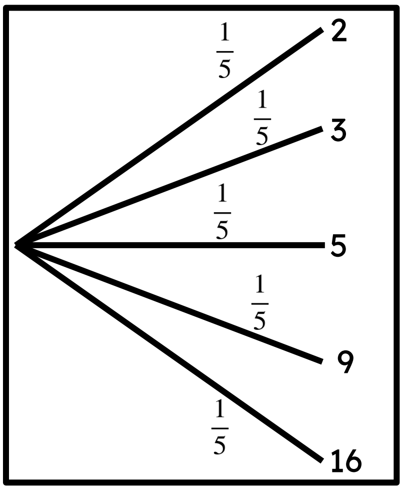
Task B: Comparing trees that show events and outcomes

1) Izzy will spin spinner A once and look for the event that the spinner lands on a prime number. Izzy constructs a **probability tree** of outcomes.

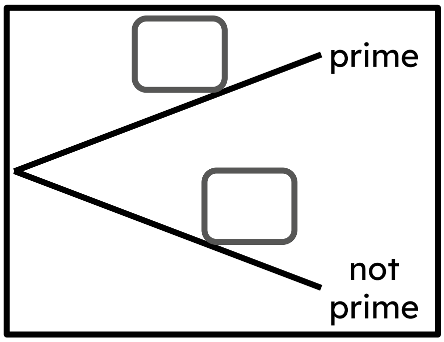
Complete the **probability tree** showing two events: landing on a prime, or landing on a non-prime.



spinner A

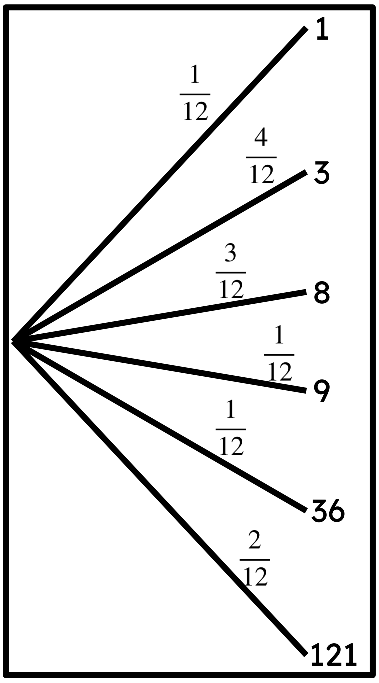


probability tree of outcomes



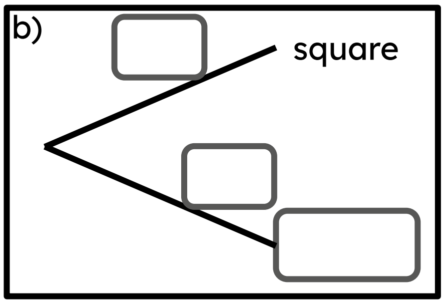
probability tree showing two events

2) Aisha creates a **probability tree** of outcomes for a trial of spinning a spinner with 12 sectors.



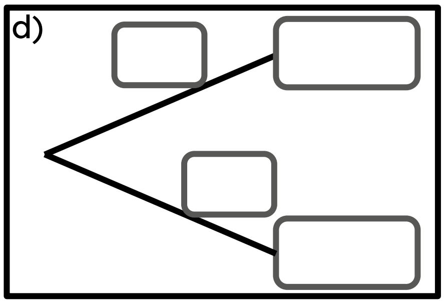
a) Which of these outcomes are square numbers?

b) Complete the **probability tree** of two events: landing on a square number, and landing on a non-square number.



c) Which of these outcomes are even numbers?

d) Complete the **probability tree** of two events, where one event is the spinner landing on an even number.



3) Lucas and Andeep play a game. Whoever gets the highest number wins, and if they get the same number, they draw.
Lucas spins the spinner, which lands on a 5.
Andeep is about to spin the spinner.

Complete this **probability tree** to show the events of who could win.